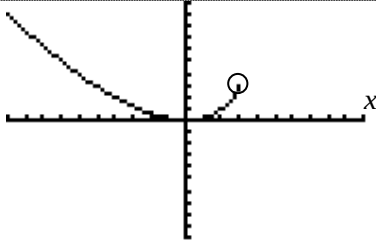


2009 HCI Preliminary Examinations H2 Mathematics Paper 2

No.	Solution
1	$\frac{dy}{dx} = \frac{du}{dx}(3-x) - u$ $(3-x) \left[\frac{du}{dx}(3-x) - u \right] = x - (3-x)u$ $\frac{du}{dx}(3-x) - u = \frac{x}{3-x} - u \Rightarrow \frac{du}{dx} = \frac{x}{(3-x)^2}$ <u>Method 1: (Partial Fractions)</u> $\frac{du}{dx} = \frac{x}{(3-x)^2} = \frac{3}{(3-x)^2} - \frac{1}{(3-x)}$ $u = \int \left(\frac{3}{(3-x)^2} - \frac{1}{(3-x)} \right) dx$ $= \frac{3}{3-x} + \ln(3-x) + c$ $y = (3-x) [\ln(3-x) + c] + 3$
	<u>Method 2: (Splitting of numerator)</u> $\frac{du}{dx} = \frac{x}{(3-x)^2} = \frac{1}{2} \left(\frac{2x-6+6}{9-6x+x^2} \right)$ $\frac{du}{dx} = \frac{1}{2} \left[\frac{2x-6}{9-6x+x^2} + \frac{6}{(3-x)^2} \right]$ $u = \frac{1}{2} \left[\ln(3-x)^2 + \frac{6}{3-x} \right] + c \quad \text{or} \quad \ln(3-x) + \frac{3}{3-x} + c$ $y = (3-x) \left[\frac{1}{2} \ln(3-x)^2 + \frac{3}{3-x} + c \right]$ $y = (3-x) [\ln(3-x) + c] + 3$
	When $x = 0, y = 0$, $3(\ln 3 + c) + 3 = 0 \Rightarrow c = -1 - \ln 3$ $y = (3-x) [\ln(3-x) - 1 - \ln 3] + 3$ As $x \rightarrow 3, 3-x \rightarrow 0 \Rightarrow y \rightarrow 0 + 3 = 3$. Graph of $y = (3-x) [\ln(3-x) - 1 - \ln 3] + 3, x < 3$. y

	
2	Given $\vec{OA} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \vec{OB} = \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}$ Equation of $l: \mathbf{r} = m \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}, m \in \mathbb{R}$. <u>Method 1:</u> Let F be the foot of perpendicular from C to the line l, $\vec{CF} \cdot \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} = 0$. Thus, $(\vec{OF} - \vec{OC}) \cdot \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -3m - k \\ 2m + 2k \\ m - k \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} = 0$ $\Rightarrow 9m + 3k + 4m + 4k + m - k = 0 \Rightarrow m = -\frac{3}{7}k$ Since F is the mid-point of C and C', $\vec{OC'} = 2\vec{OF} - \vec{OC}$ $= \frac{1}{7}k \begin{pmatrix} 11 \\ 2 \\ -13 \end{pmatrix} \text{ (Shown)}$ <u>Method 2:</u> Let $\vec{OF}$ be the projection vector of $\vec{OC}$ on the line l. $\text{Then } \vec{OF} = \left(\frac{\vec{OC} \cdot \vec{OB}}{\vec{OB} \cdot \vec{OB}} \right) \vec{OB} = \frac{k \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}}{\sqrt{14} \sqrt{14}}$

	$= -\frac{3}{7}k \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}$ Since F is the mid-point of C and C', using Ratio Theorem Since F is the mid-point of C and C', $\overrightarrow{OC'} = 2\overrightarrow{OF} - \overrightarrow{OC}$ $= -\frac{6}{7}k \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} - k \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \frac{1}{7}k \begin{pmatrix} 11 \\ 2 \\ -13 \end{pmatrix} \quad (\text{shown})$
	When BC is perpendicular to OA, $\overrightarrow{BC} \cdot \overrightarrow{OA} = 0$ $\Rightarrow \left[\begin{pmatrix} k \\ -2k \\ k \end{pmatrix} - \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = 0$ $\Rightarrow \begin{pmatrix} k+3 \\ -2k-2 \\ k-1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = 0$ $\Rightarrow k+3-4k-4-3k+3=0$ $\Rightarrow k = \frac{1}{3}$

	When $k = \frac{1}{3}$, $\vec{OC} = \begin{pmatrix} 1/3 \\ -2/3 \\ 1/3 \end{pmatrix}$ and $\vec{OC'} = \frac{1}{21} \begin{pmatrix} 11 \\ 2 \\ -13 \end{pmatrix}$. Hence, $\text{LHS} = 6\vec{OA} + 15\vec{OC} = 6 \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + 15 \begin{pmatrix} 1/3 \\ -2/3 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 11 \\ 2 \\ -13 \end{pmatrix} \quad (\text{Verified})$ $\text{RHS} = 21\vec{OC'} = \begin{pmatrix} 11 \\ 2 \\ -13 \end{pmatrix} = \text{LHS}$ <u>Method 1:</u> $6\vec{OA} + 15\vec{OC} - 21\vec{OC'} = \mathbf{0}$ $\Rightarrow 6\vec{OA} - 6\vec{OC'} + 15\vec{OC} - 15\vec{OC'} = \mathbf{0}$ $\Rightarrow 6(\vec{OA} - \vec{OC'}) + 15(\vec{OC} - \vec{OC'}) = \mathbf{0}$ $\Rightarrow 6\vec{C'A} = 15\vec{CC'}$ Thus, $C'A$ is parallel to CC'. Since C' is the common point, the points A, C and C' are collinear. <u>Method 2:</u> $6\vec{OA} + 15\vec{OC} = 21\vec{OC'}$ $\Rightarrow \vec{OC'} = \frac{6\vec{OA} + 15\vec{OC}}{21}$ Hence by Ratio Theorem, we can deduce that A, C and C' are collinear.
3(i)	Sub $w = x + yi$ into the first equation to obtain: $x^2 + y^2 - 16\sqrt{3}i + 8i(x + yi) = 0$ Imaginary parts: $-16\sqrt{3} + 8x = 0 \Rightarrow x = 2\sqrt{3}$ Real parts: $x^2 + y^2 - 8y = 0$ Sub $x = 2\sqrt{3}$: $y^2 - 8y + 12 = 0$ $(y - 6)(y - 2) = 0 \Rightarrow y = 2 \text{ or } 6 \quad (\text{NA } \because y < 5)$ $w = 2\sqrt{3} + 2i = 2(\sqrt{3} + i)$

(ii)	<u>Method 1:</u> $w = 2(\sqrt{3} + i) = 4 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \left(\text{or } w = 4e^{i\frac{\pi}{6}} \right)$ $w^n = 4^n \left(\cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6} \right) \text{ is real if } \sin \frac{n\pi}{6} = 0.$ $\frac{n\pi}{6} = k\pi \text{ where } k \in \mathbb{Z}.$ $w^n \text{ is real provided } n = 6k \text{ where } k \in \mathbb{Z}.$
	<u>Method 2:</u> $\arg(w) = \frac{\pi}{6} \Rightarrow \arg(w^n) = \frac{n\pi}{6}$ Since w^n is real, $\frac{n\pi}{6} = k\pi$ $n = 6k \text{ where } k \in \mathbb{Z}.$
(iii)	$1 + \left(\frac{w}{4}\right)^3 + \left(\frac{w}{4}\right)^6 + \left(\frac{w}{4}\right)^9 + \dots + \left(\frac{w}{4}\right)^{21}$ $= \frac{\left(\left(\frac{w}{4}\right)^3\right)^8 - 1}{\left(\frac{w}{4}\right)^3 - 1} = \frac{\left(e^{i\frac{\pi}{6}}\right)^{24} - 1}{e^{i\frac{\pi}{2}} - 1} = \frac{e^{4\pi i} - 1}{i - 1} = 0$
(iv)	$\arg\left(\frac{zi}{1+i}\right) = \arg(z) + \arg i - \arg(1+i) = \frac{3}{4}\pi$ $\Rightarrow \arg(z) = \frac{\pi}{2} \text{ since } \arg i - \arg(1+i) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$ $\arg(z) = \frac{\pi}{2} \text{ \& } z = 4 \Rightarrow z = 4i$
(v)	<u>Method 1:</u> Area of triangle required = $\frac{1}{2}$ base x height = $\frac{1}{2}(4)(2\sqrt{3}) = 4\sqrt{3} \text{ unit}^2$ <u>Method 2 (Remove from students' solution):</u> $\angle ZOW = \frac{\pi}{3} \text{ (using } \arg z - \arg w = \frac{\pi}{2} - \frac{\pi}{6})$ Area of triangle required =

	$\frac{1}{2}(OZ)(OW)\sin \angle ZOW = \frac{1}{2}(4)(4)\sin \frac{\pi}{3} = 4\sqrt{3} \text{ unit}^2$
4	$f(x) = e^{\sin x} = e^{x - \frac{x^3}{3!} + \dots}$ $= 1 + \left(x - \frac{x^3}{3!}\right) + \frac{\left(x - \frac{x^3}{3!}\right)^2}{2!} + \frac{\left(x - \frac{x^3}{3!}\right)^3}{3!} + \frac{\left(x - \frac{x^3}{3!}\right)^4}{4!} + \dots$ $= 1 + x - \frac{x^3}{6} + \frac{1}{2}\left(x^2 - \frac{2x^4}{6}\right) + \frac{x^3}{6} + \frac{x^4}{24} + \dots$ $= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \dots$
	$e^{\sin(x+\pi)} = e^{\sin x \cos \pi + \cos x \sin \pi} = e^{-\sin x} = e^{\sin(-x)}$ $= 1 - x + \frac{x^2}{2} - \frac{x^4}{8} + \dots \quad [\text{Replace } x \text{ by } -x]$
(i)	$4 \frac{dy}{dx} = (y+1)^2$ Differentiating with respect to x, $4 \frac{d^2y}{dx^2} = 2(y+1) \frac{dy}{dx} \Rightarrow 2 \frac{d^2y}{dx^2} = (y+1) \frac{dy}{dx} \text{ --- (1)}$ $2 \frac{d^3y}{dx^3} = (y+1) \frac{d^2y}{dx^2} + \frac{dy}{dx} \frac{dy}{dx} = (y+1) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 \text{ --- (2)}$ Sub $x = 0, y = 1$, $2 \frac{d^4y}{dx^4} = \frac{dy}{dx} \frac{d^2y}{dx^2} + (y+1) \frac{d^3y}{dx^3} + 2 \frac{dy}{dx} \frac{d^2y}{dx^2}$ $4 \frac{dy}{dx} = (1+1)^2 \Rightarrow \frac{dy}{dx} = 1$ $2 \frac{d^2y}{dx^2} = (1+1)1 = 2 \Rightarrow \frac{d^2y}{dx^2} = 1$ $2 \frac{d^3y}{dx^3} = (1+1)1 + 1^2 \Rightarrow \frac{d^3y}{dx^3} = \frac{3}{2}$ $2 \frac{d^4y}{dx^4} = 1 + (1+1) \frac{3}{2} + 2 \Rightarrow \frac{d^4y}{dx^4} = 3$ Using the Maclaurin's formula, $g(x) = 1 + x + \frac{x^2}{2!} + \frac{3x^3}{2 \cdot 3!} + \frac{3x^4}{4!} + \dots$

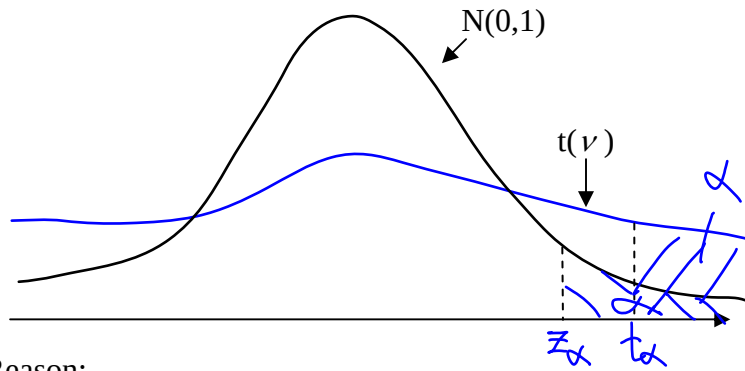
	$= 1 + x + \frac{x^2}{2} + \frac{x^3}{4} + \frac{x^4}{8} + \dots$ $g(x) - f(x) = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{4} + \frac{x^4}{8} + \dots \right) - \left(1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \dots \right)$ $\approx \frac{x^3 + x^4}{4}$						
(ii)	As $x \rightarrow 0$, $g(x) - f(x) \approx \frac{x^3 + x^4}{4} \rightarrow 0$. When x is close 0, the graph of $y = g(x)$ is close to the graph of $y = f(x)$ OR the graph of $y = g(x)$ is a good approximation to the graph of $y = f(x)$.						
(iii)	Estimated error $= \int_{-k}^k \frac{x^3 + x^4}{4} dx = \left[\frac{1}{4} \left(\frac{x^4}{4} + \frac{x^5}{5} \right) \right]_{-k}^k = \frac{k^5}{10}$ $\frac{k^5}{10} < 10^{-2} \Rightarrow k^5 < 10^{-1} \Rightarrow k < 0.631 \Rightarrow 0 < k < 0.631$.						
5(i)	Let X denote the number of emergency telephone boxes in 1 km ² . Then $X : \text{Po}(0.6)$.						
(ii)	Poisson distribution model is not suitable since the average rate of occurrence of the emergency telephone boxes in the rural area (including the city) is not constant throughout this area as city and rural area have different density for distribution of emergency telephone boxes						
6a)	No. Because not every resident will visit the shopping mall, or Consider total population size, n, and take sample interval to be $\frac{n}{50}$ instead of choosing every 10th shopper (or any other equivalent / logical reason).						
(b)	The surveyor is given quotas to fill. An example is the table shown below: <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <th colspan="2">Gender</th></tr> <tr> <th>Male</th><th>Female</th></tr> <tr> <td>25</td><td>25</td></tr> </table> The surveyor is given a free choice in picking the shoppers to fill the quotas.	Gender		Male	Female	25	25
Gender							
Male	Female						
25	25						
(c)	Increase sample size. Interview people from different shopping malls. Increase no. of categories or strata.						
7	Case 1: Only 1 female does not know hip-hop Prob $= \frac{{}^{10}C_2 \times {}^8C_1 \times {}^{12}C_1}{{}^{20}C_2 \times {}^{20}C_2}$ or $\frac{10}{20} \times \frac{9}{19} \times \frac{8}{20} \times \frac{12}{19} \times 2$ $= 0.119668$ or $\frac{216}{1805}$ Case 2: Only 1 male does not know hip-hop						

	$\text{Prob} = \frac{{}^8C_2 \times {}^{10}C_1 \times {}^{10}C_1}{{}^{20}C_2 \times {}^{20}C_2} \text{ or } \frac{8}{20} \times \frac{7}{19} \times \frac{10}{20} \times \frac{10}{19} \times 2$ $= 0.0775623 \text{ or } \frac{28}{361}$ Case 3: All 4 people know hip-hop $\text{Prob} = \frac{{}^{10}C_2 \times {}^8C_2}{{}^{20}C_2 \times {}^{20}C_2} \text{ or } \frac{10}{20} \times \frac{9}{19} \times \frac{8}{20} \times \frac{7}{19}$ $= 0.034903 \text{ or } \frac{63}{1805}$ $P(A) = \frac{216}{1805} + \frac{28}{361} + \frac{63}{1805} = \frac{419}{1805} \text{ or } 0.232$
	$P(B A) = P(\text{all 4 married} \text{at least 3 know hip-hop})$ $= \frac{P(\text{all 4 married} \cap \text{at least 3 know hip-hop})}{P(\text{at least 3 know hip-hop})}$ $= \frac{P(\text{all 4 married} \cap \text{all 4 know hip-hop})}{P(\text{at least 3 know hip-hop})}$ $= \frac{P(\text{all 4 married})}{P(\text{at least 3 know hip-hop})}$ $= \frac{\left(\frac{{}^2C_2 \times {}^4C_2}{{}^{20}C_2 \times {}^{20}C_2} \right)}{\left(\frac{419}{1805} \right)} \text{ or } \frac{\left(\frac{2}{20} \times \frac{1}{19} \times \frac{4}{20} \times \frac{3}{19} \right)}{\left(\frac{419}{1805} \right)} = 0.000716$
	$P(B) = \frac{{}^2C_2 \times {}^4C_2}{{}^{20}C_2 \times {}^{20}C_2} \text{ or } \frac{2}{20} \times \frac{1}{19} \times \frac{4}{20} \times \frac{3}{19} = 0.000166$ Since $P(B A) \neq P(B)$, A and B are not independent events.

8	Assumption: The total no. of tickets in the box has to be large so that the probability of getting a winning ticket is approximately constant after each ticket is drawn. (i) Let X be the number of winning tickets in a draw of 5 tickets. $X \sim B(5, \frac{1}{6})$ $P(X \geq 1) = 1 - P(X = 0) = 0.5981224$ <u>Method 1:</u> Required probability = $3 \times P(X \geq 1)[P(X = 0)]^2$ $= 3 \times (0.5981224)(0.4018776)^2$ $= 0.2897996 \approx 0.290$ <u>Method 2:</u> Let Y be the number of customers out of 3 with at least one winning ticket. $Y \sim B(3, 0.5981224)$ Required probability = $P(Y = 1) = 0.2897996 \approx 0.29$
ii)	Since $n = 60$ is large, by the Central Limit Theorem, $\bar{X} \sim N(\frac{5}{6}, \frac{5}{432}) \text{ approximately}$ Required probability = $P(\bar{X} \geq 1) = 0.0607$ The event that all 60 customers each obtains at least one winning ticket is a subset of the event that the average no. of winning tickets obtained by the 60 customers is at least one. Hence the probability is smaller.
9i)	Case 1: I _ _ _ _ _ I: $\text{No. of ways} = \frac{9!}{2! 2!} = 90\,720$ Case 2: I _ _ _ _ _ O or O _ _ _ _ _ I: $\text{No. of ways} = 2 \times \frac{9!}{2! 2! 2!} = 90\,720$ $\therefore$ total no. of ways = 181 440
(ii)	$\text{No. of arrangements} = \frac{11!}{3! 2! 2!} - \frac{8!}{2! 2!}$ $= 1\,663\,200 - 10\,080$ $= 1\,653\,120$
(iii)	The first letter 'T' can be placed at the 1st, 2nd, 3rd, ..., 7th position. The second 'T' is then placed 4 positions after the first 'T'. $\text{No. of ways} = 7 \times \frac{9!}{3! 2!} - 1 = 211\,679$ (case where it is correct word is excluded)

	$s^2 = \frac{1}{99} [36\,990\,985 - \frac{60005^2}{100}] = 9949.340909$ Hence, $Z = \frac{\bar{X} - m}{\sqrt{9949.340909/100}} \sim N(0,1)$ To reject H_0 at 5% level of significance, $\frac{600.05 - m}{\sqrt{9949.340909/100}} \geq 1.644853626$ $\Rightarrow m \leq 584$ (3s.f.)
b(i)	If the claim that the mean SAT score has not improved is in fact true, 100p% of tests using same method will lead to wrongly accepting the claim that the mean SAT score has improved.
(ii)	t-test should be carried out because sample size = 10 is small and population variance is not given. Assumptions: The SAT score of students in the school follows a normal distribution. The SAT scores of students are independent (or randomly selected) from one another.

- (iii) Not necessarily true that the same conclusion as t -test is reached (i.e., the null hypothesis is not rejected) if z -test is carried out.



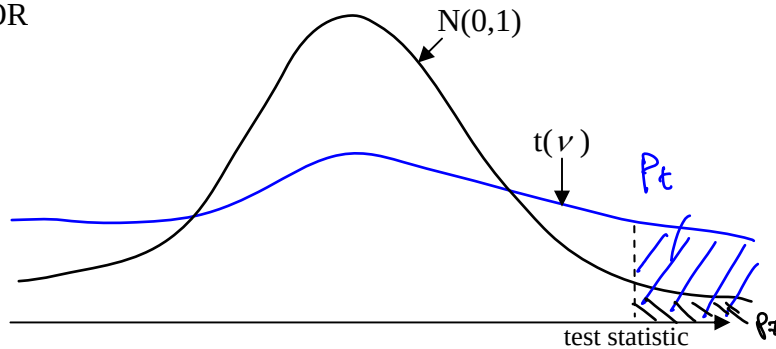
Reason:

For a fixed level of significance level α , $z_\alpha < t_\alpha$ (refer to graph above).

When H_0 is not rejected for a t -test, there are two possible conclusions for a z -test.

- 1) $z_\alpha < \text{test statistic} < t_\alpha$: H_0 will be rejected if z -test is used.
- 2) $\text{test statistic} < z_\alpha < t_\alpha$: H_0 will not be rejected if z -test is used.

OR



Let p_t and p_z be the p -values obtained using a t -test and z -test respectively.

Note that for the same test statistic, $p_t > p_z$

Since H_0 is not rejected for a t -test, $p_t > \frac{\alpha}{100}$. There are two possible conclusions for a z -test.

- 1) $p_t > \frac{\alpha}{100} \geq p_z$: H_0 will be rejected if z -test is used.
- 2) $p_z > p_t > \frac{\alpha}{100}$: H_0 will not be rejected if z -test is used.

12(i)	Let T be the travel time taken by Andy to reach the pick-up point. $T \sim N(18, 4^2)$ $P(\text{Andy takes bus B}) = P(15 < T \leq 25)$ $= 0.733313 \approx 0.733$
(ii)	$P(\text{Andy misses the buses}) = P(T > 25) = 0.040059$ Let X be the number of times in 8 work weeks that Andy misses the bus. $X \sim B(40, 0.040059)$ Since n is large and $np = 1.60236 < 5$, $X \sim \text{Po}(1.60236)$ Required probability $= P(X \geq 4)$ $= 0.0791$
(iii)	Let $\bar{T}_2$ be the average time spent on travelling to the pick-up point on the first two days of the week and $\bar{T}_3$ be the average time spent on travelling to the pick-up point on the last three days of the week. $\bar{T}_2 \sim N(18, \frac{4^2}{2})$ and $\bar{T}_3 \sim N(18, \frac{4^2}{3})$ $\bar{T}_2 - \bar{T}_3 \sim N(0, \frac{40}{3})$ $P(0 < \bar{T}_2 - \bar{T}_3 < 5) = 0.414548$ ≈ 0.415
(iv)	Let the number of minutes that Andy has to leave the home earlier be a minutes. $P(T \leq 15 + a) \geq 0.90$ or $P(T \leq k) \geq 0.90$ From GC, $P(T \leq 23.126) = 0.90$ Hence $15 + a \geq 23.126$ $\Rightarrow a \geq 8.126$ Thus he should leave his home at least 9 mins earlier. i.e his latest time to leave home is 6.51 a.m.